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(54) **LOCKING SOFFIT**

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(57) **ABSTRACT**

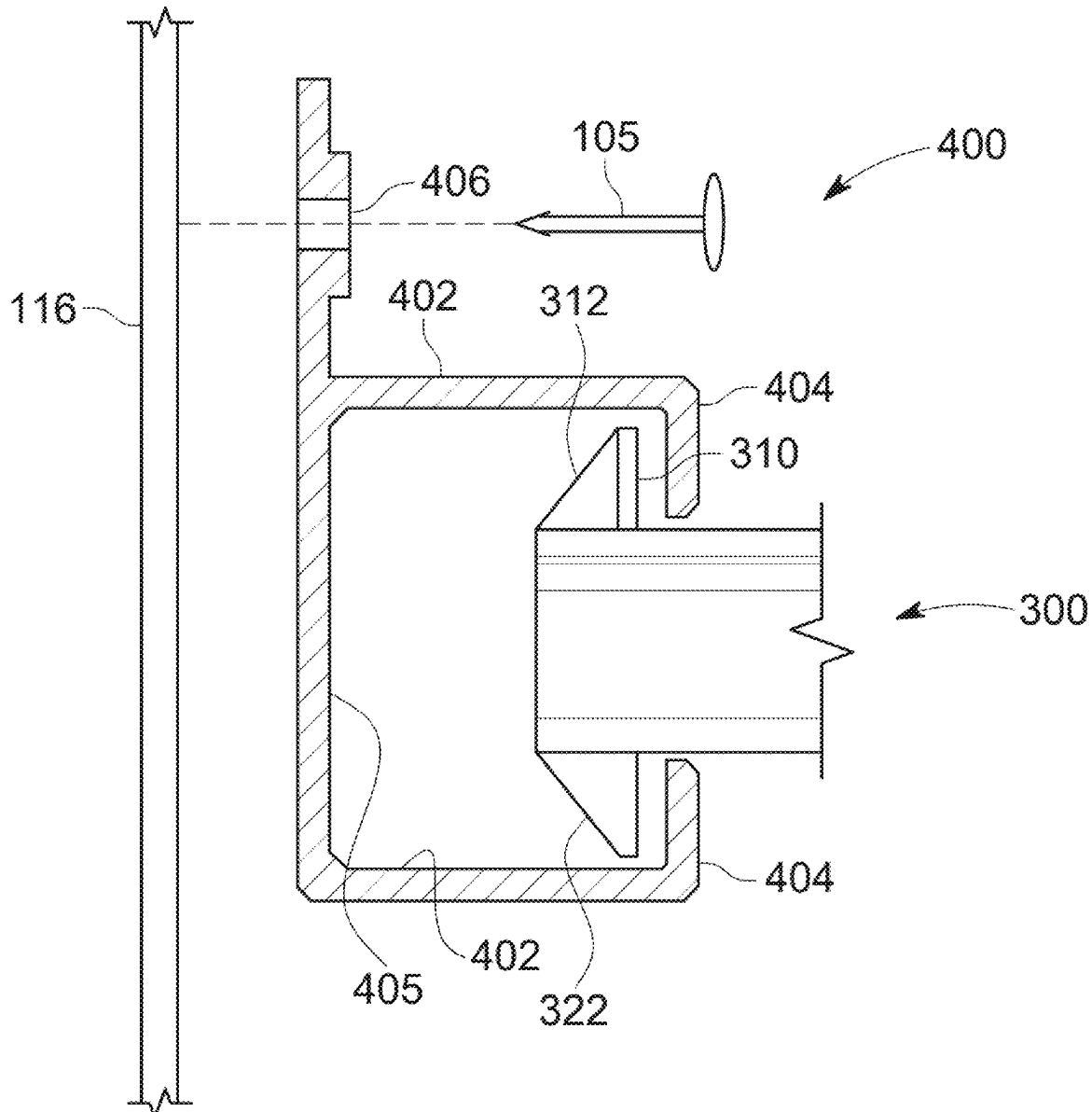
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A system for securing an end of a soffit panel to a structure includes an F-channel that is affixed to the structure. The F-channel includes a mechanism for locking an end of the soffit panel that has an upper hook and a lower hook such that after inserting the end of the soffit panel into the mechanism for locking, the upper hook and the lower hook locks the end of the soffit panel within the F-channel.





(Prior Art)

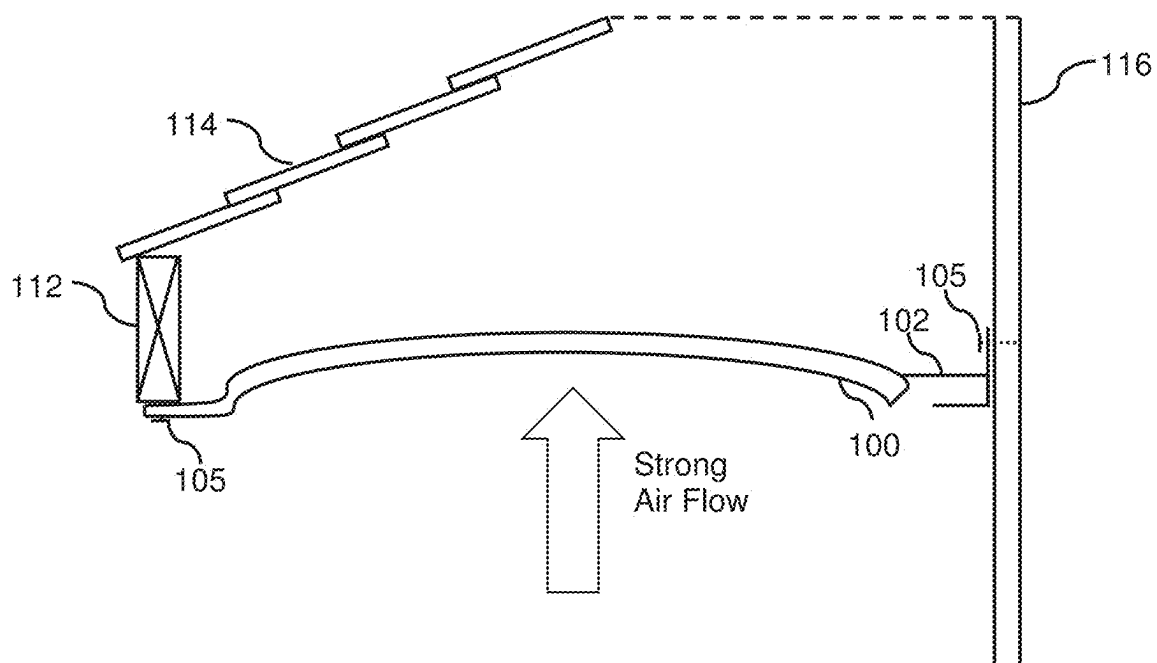


FIG. 2
(Prior Art)

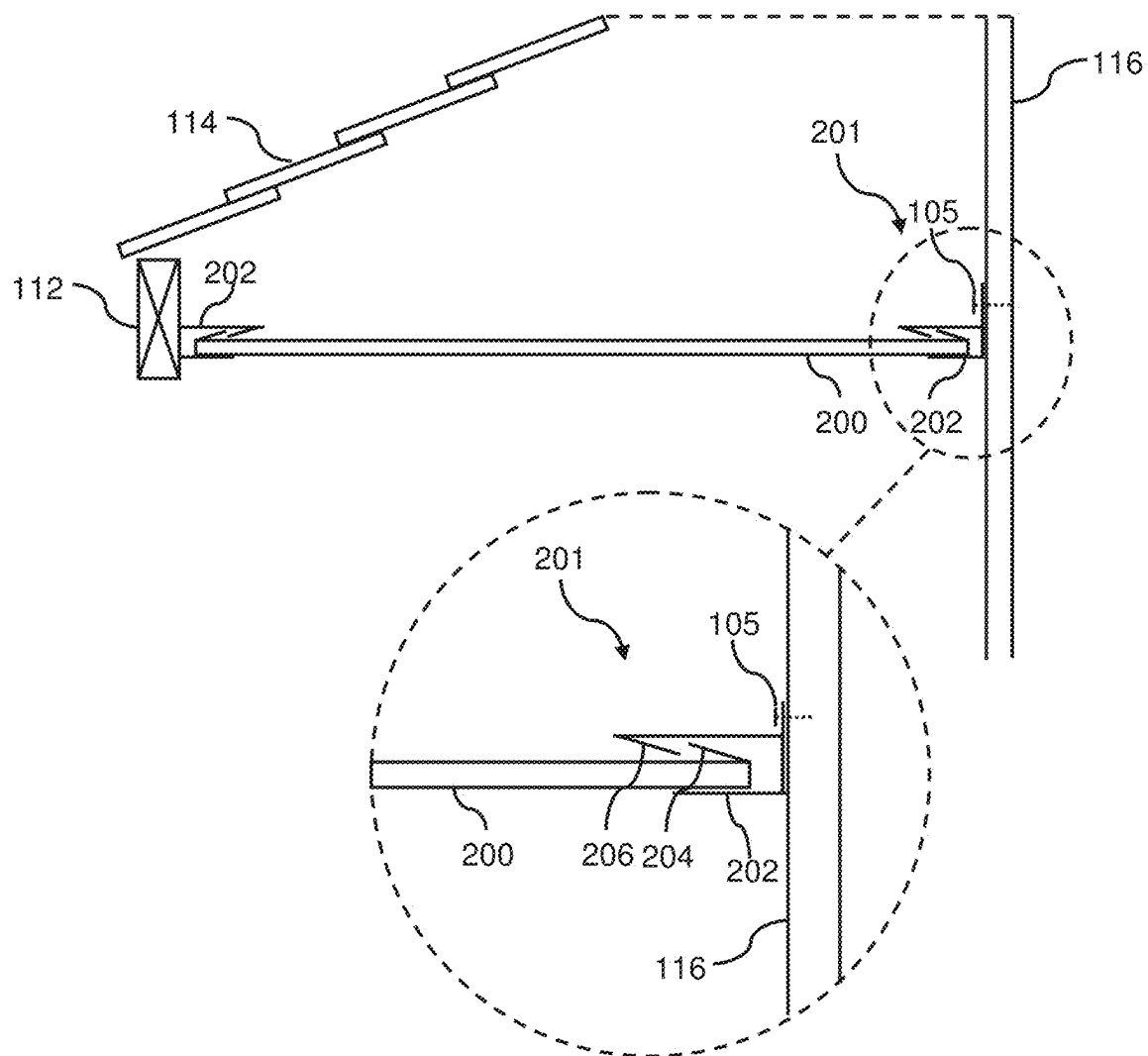


FIG. 3

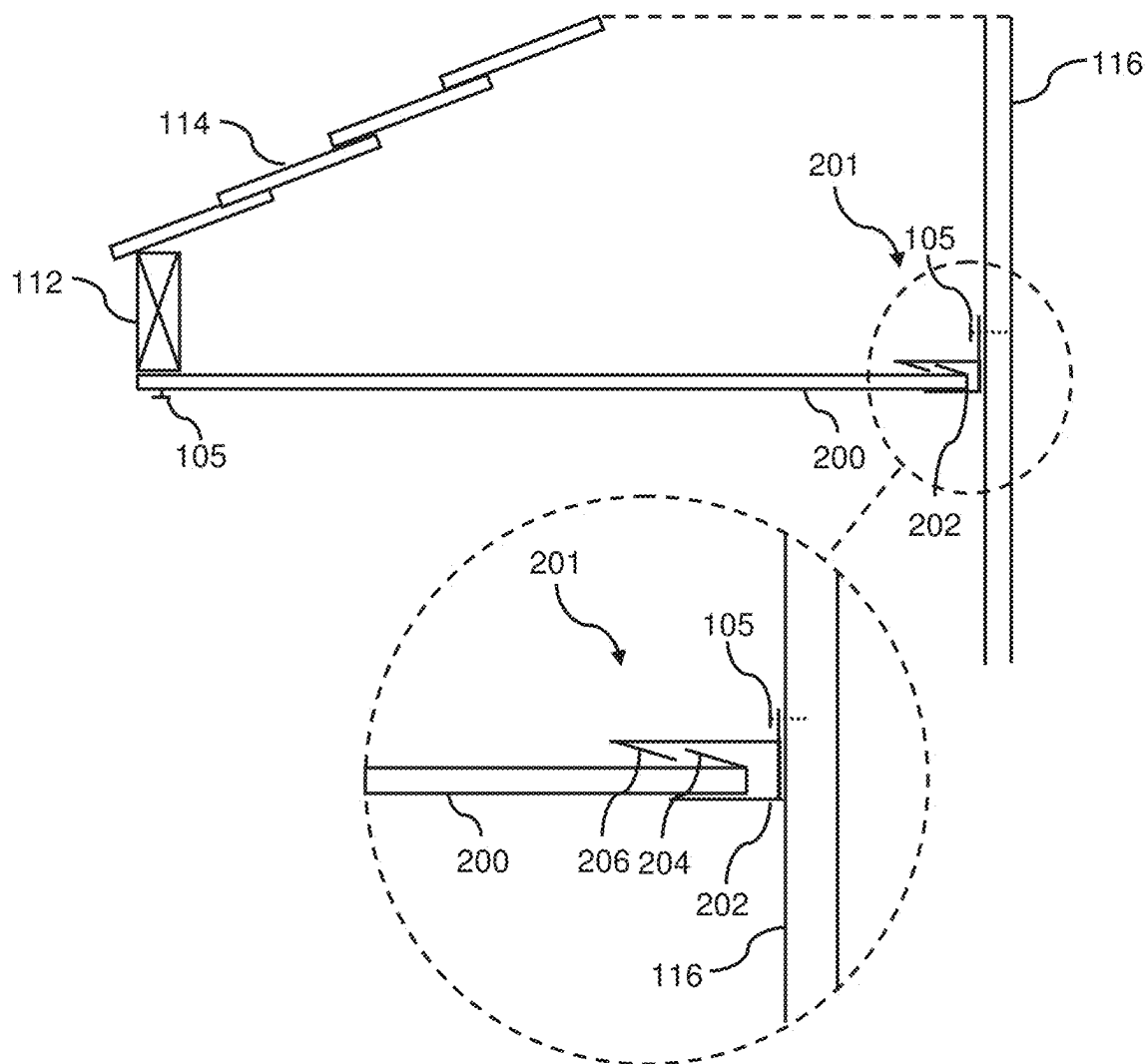


FIG. 3A

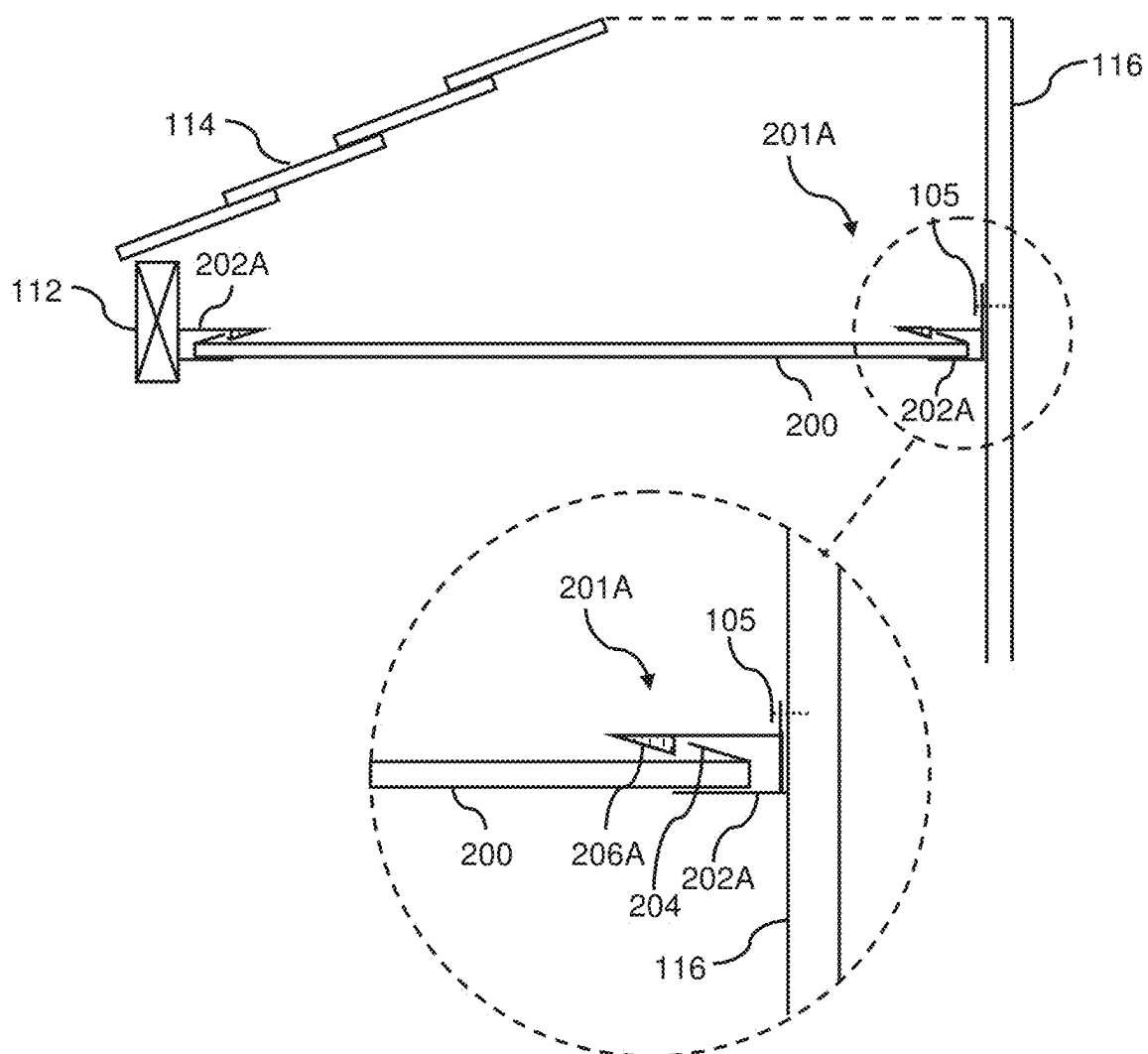


FIG. 4

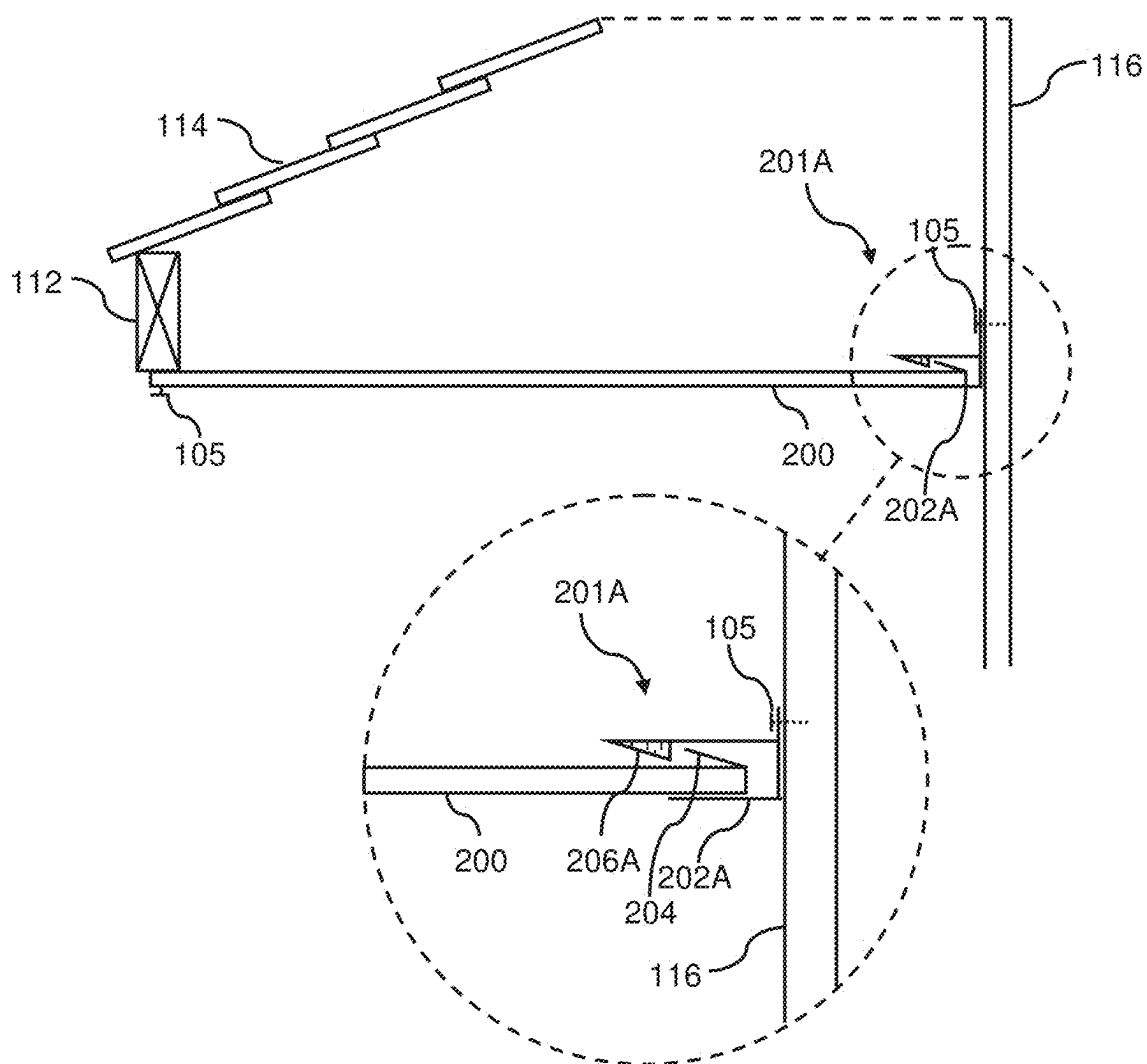


FIG. 4A

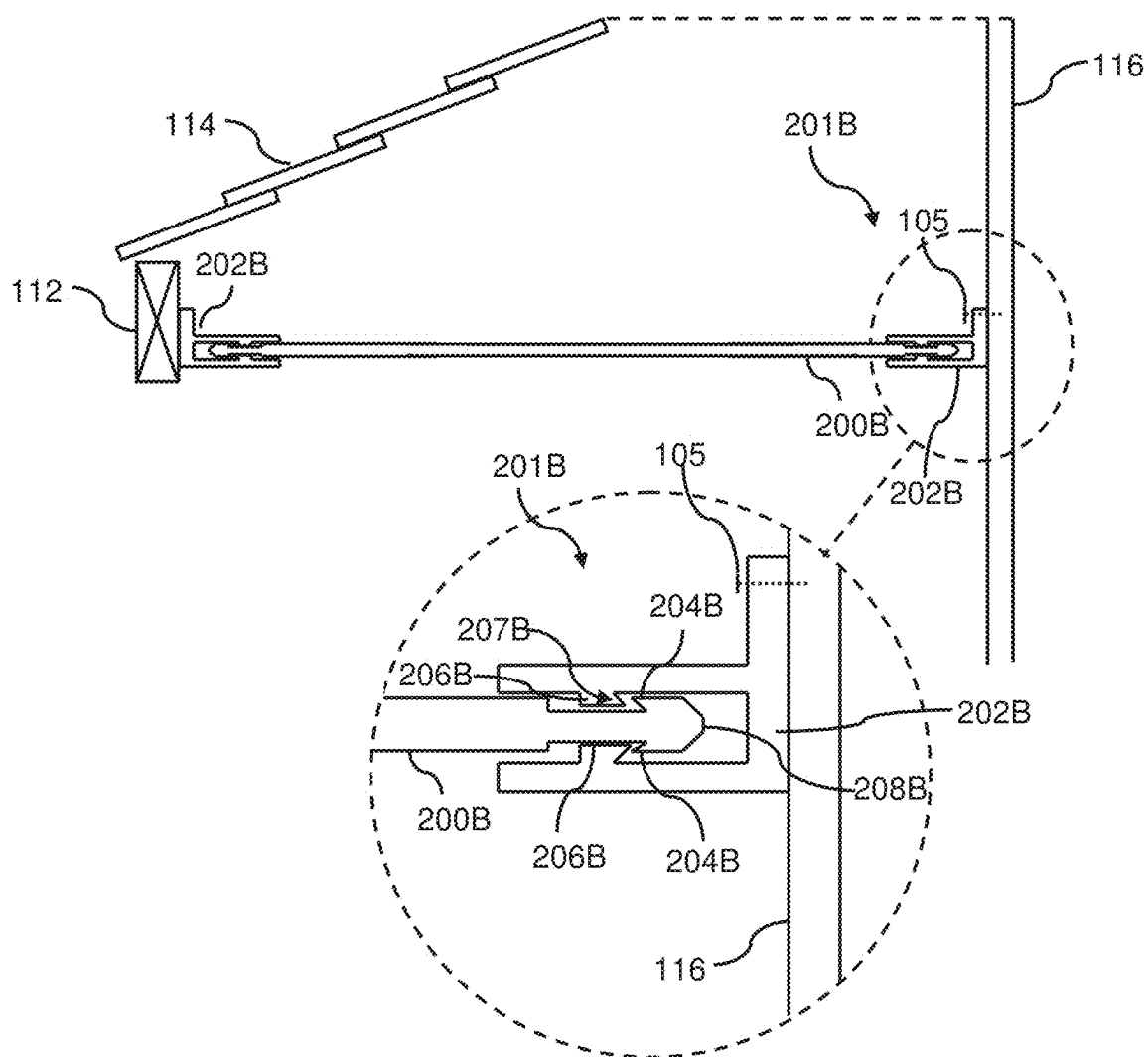


FIG. 5

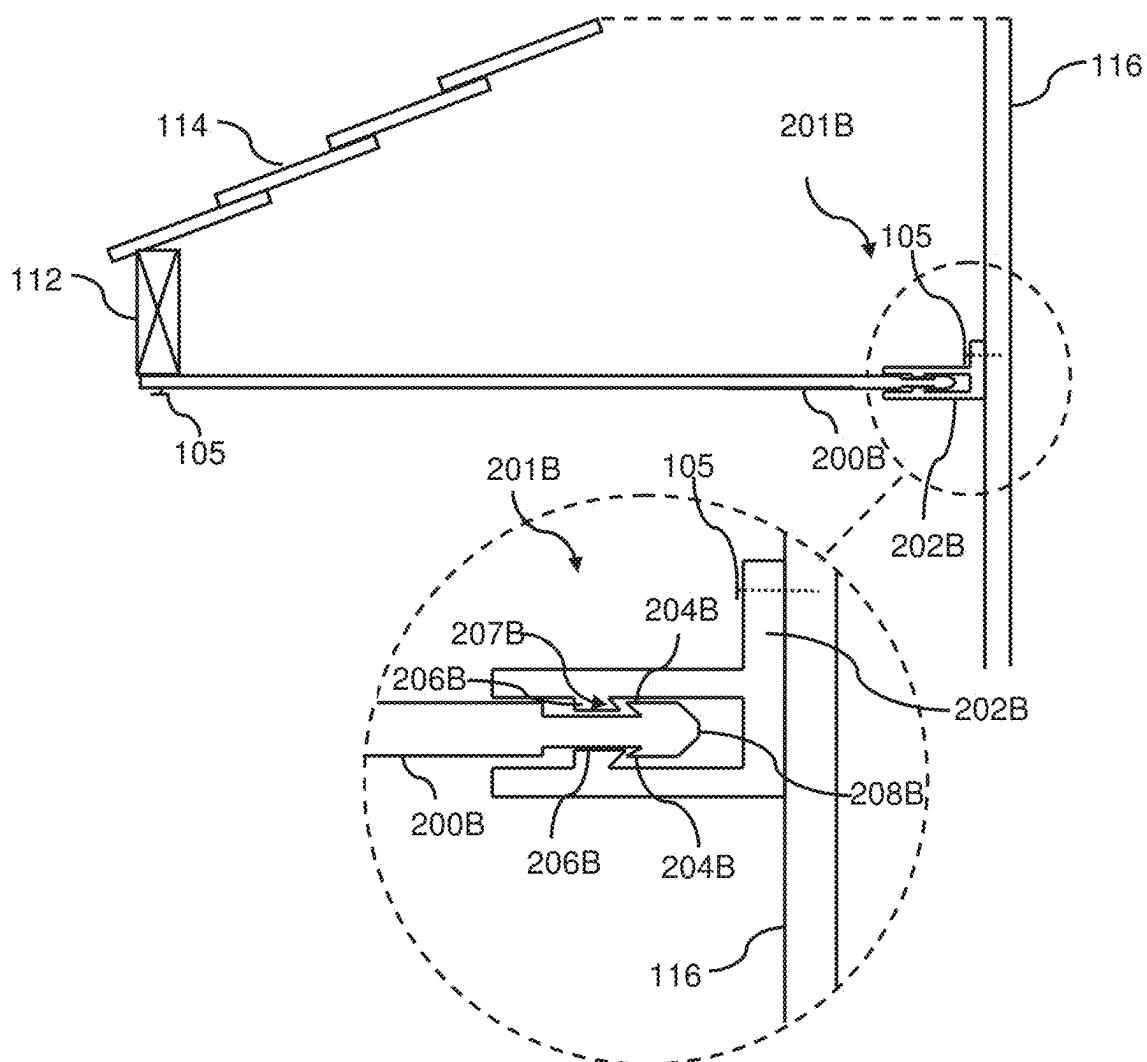


FIG. 5A

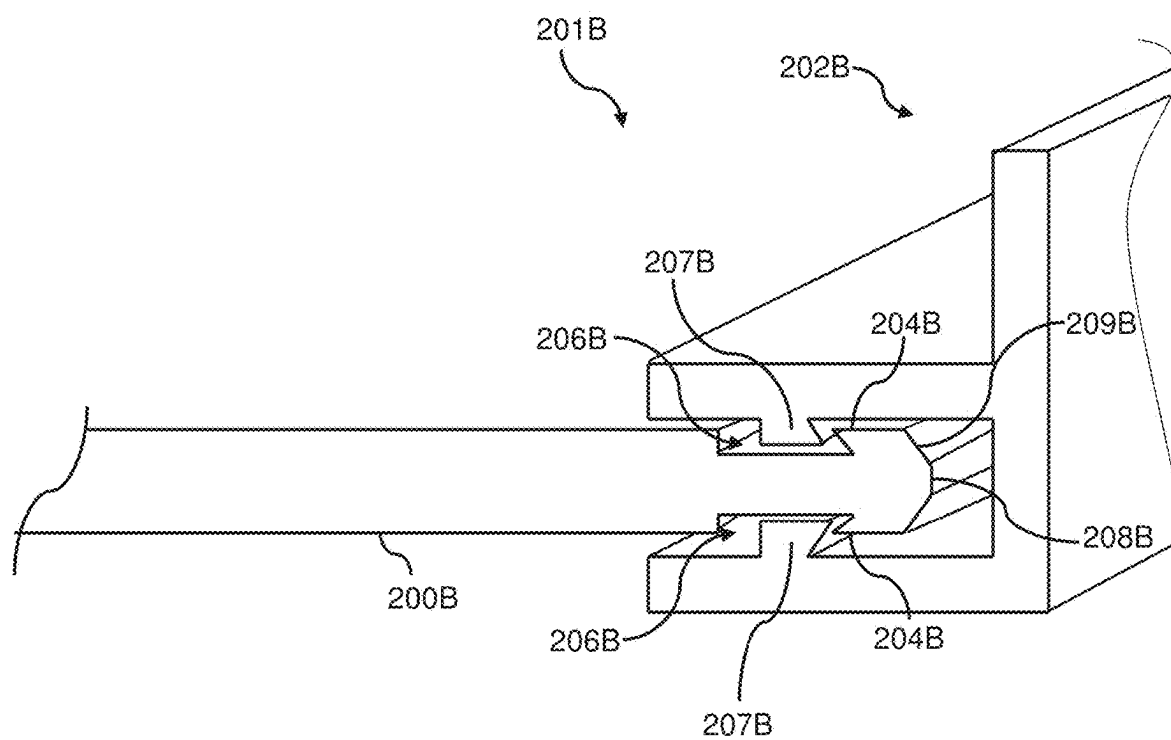


FIG. 6

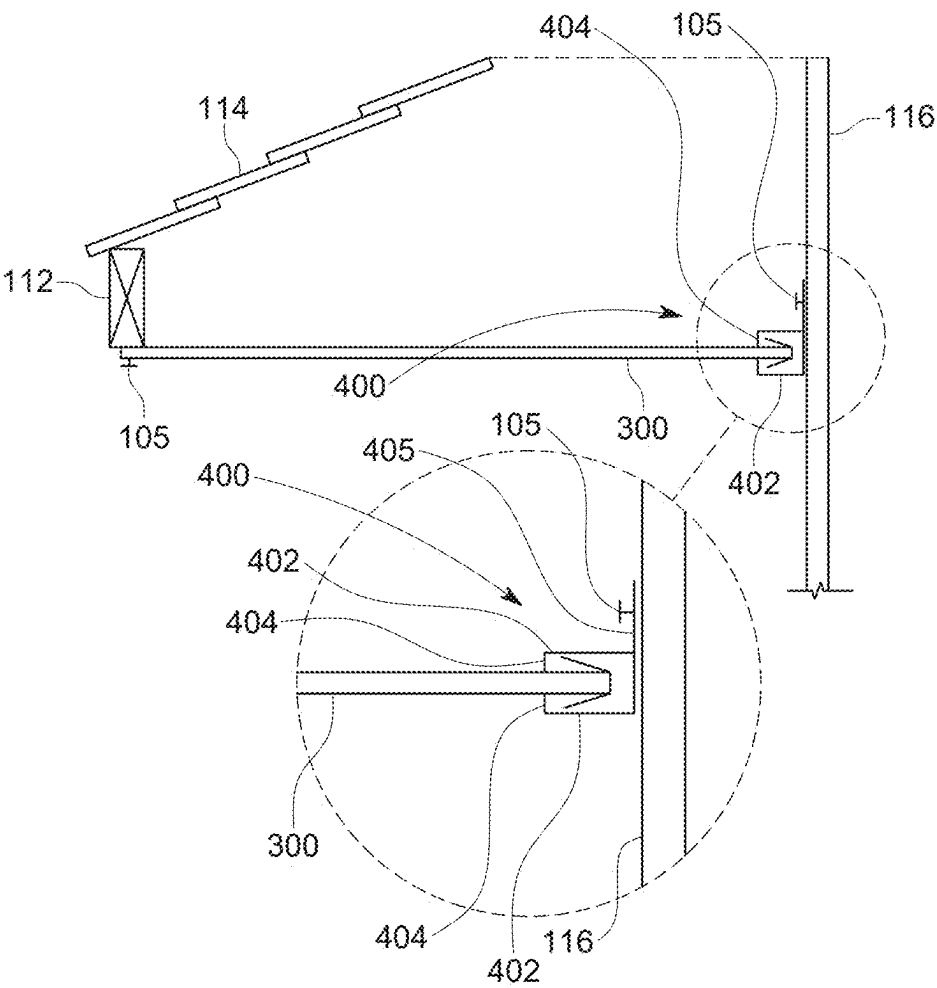


FIG. 7

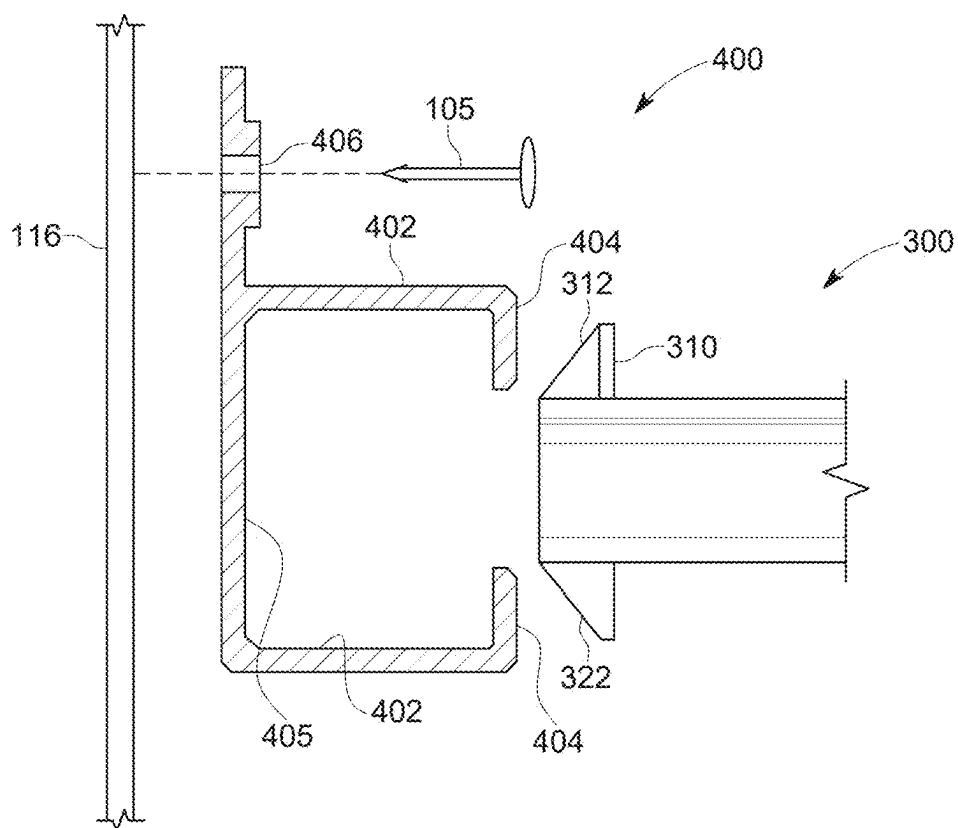


FIG. 8

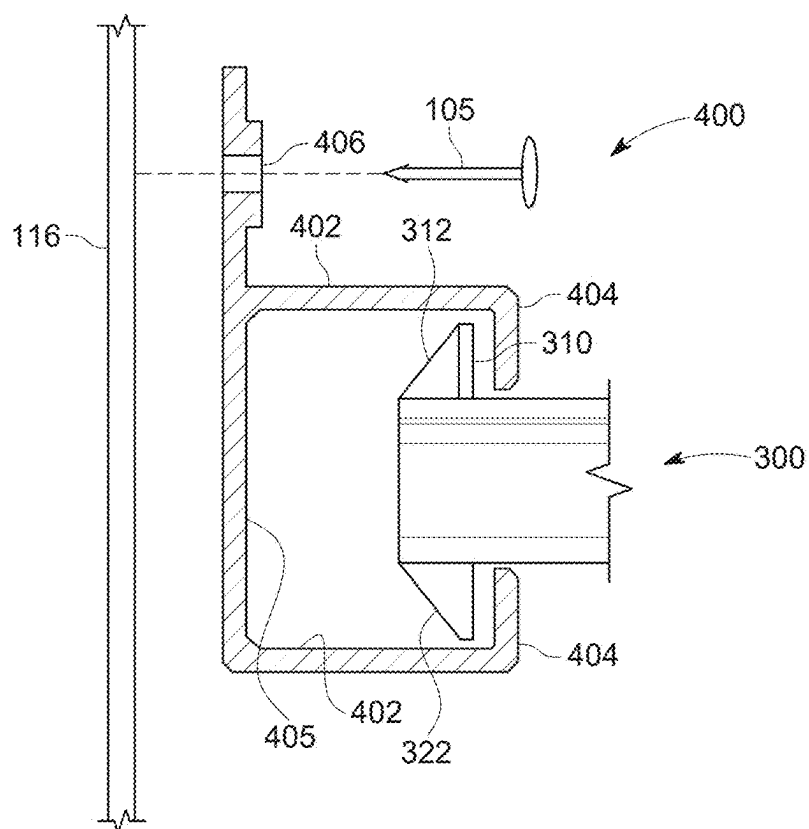


FIG. 8A

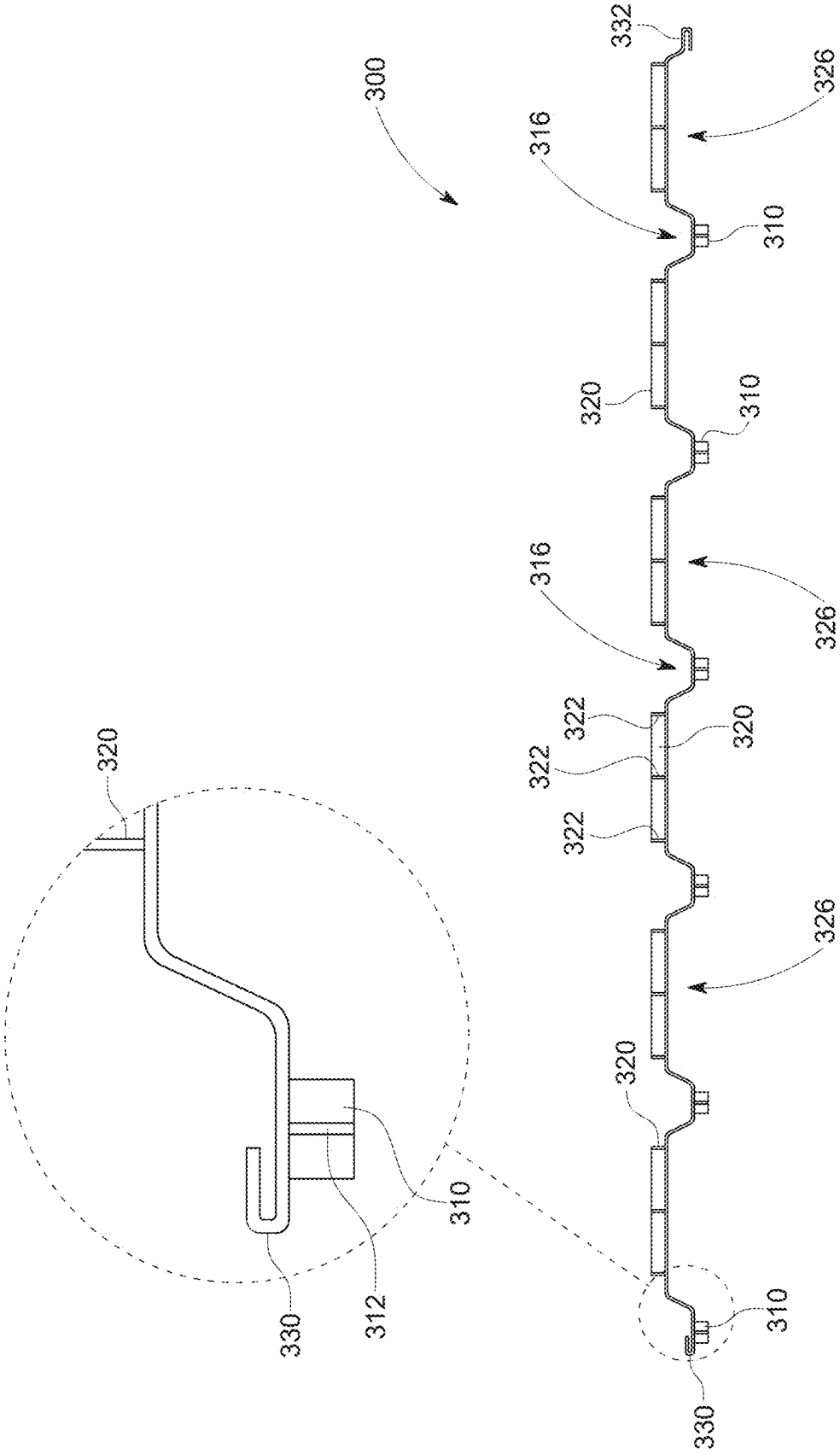


FIG. 9

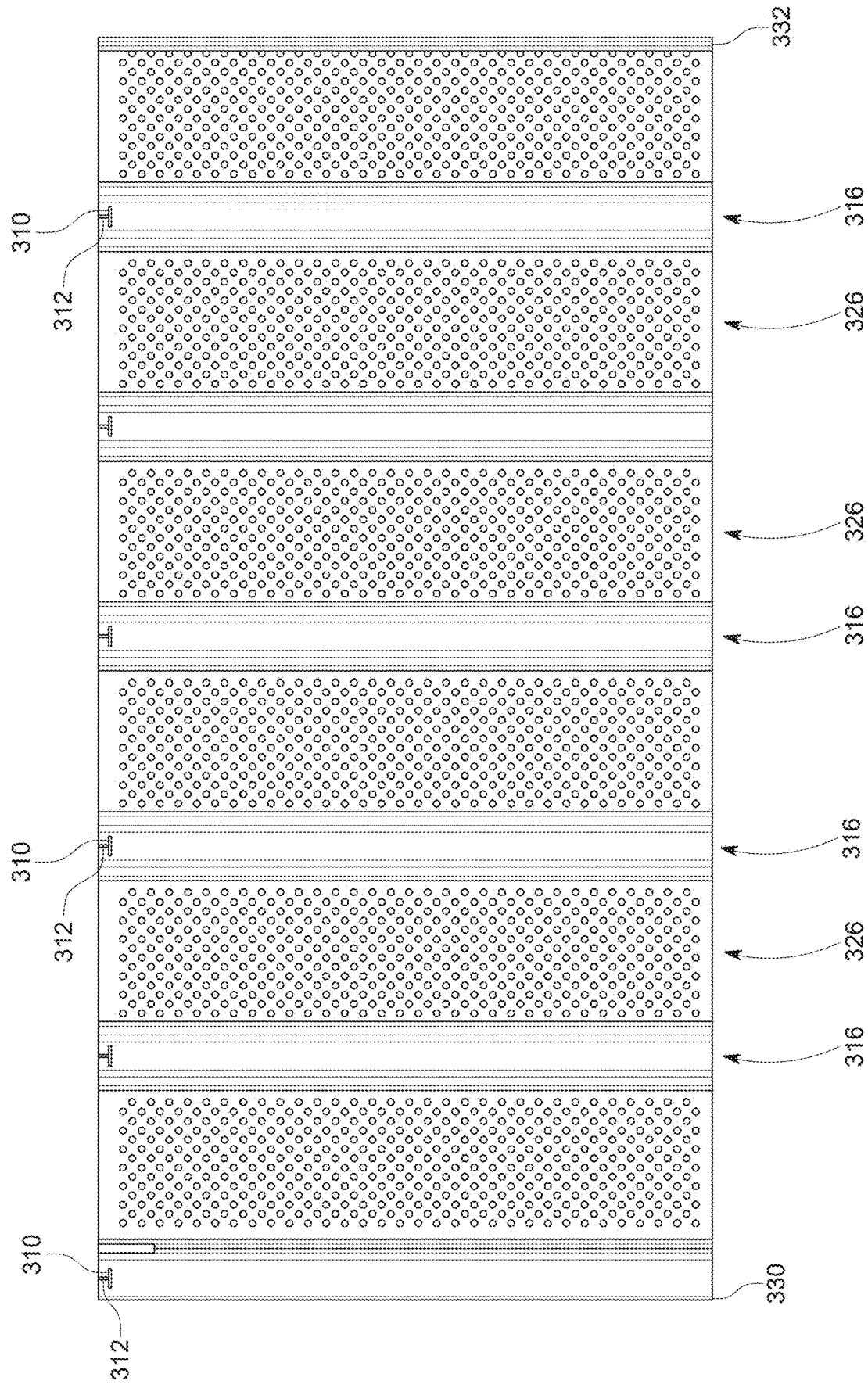


FIG. 10

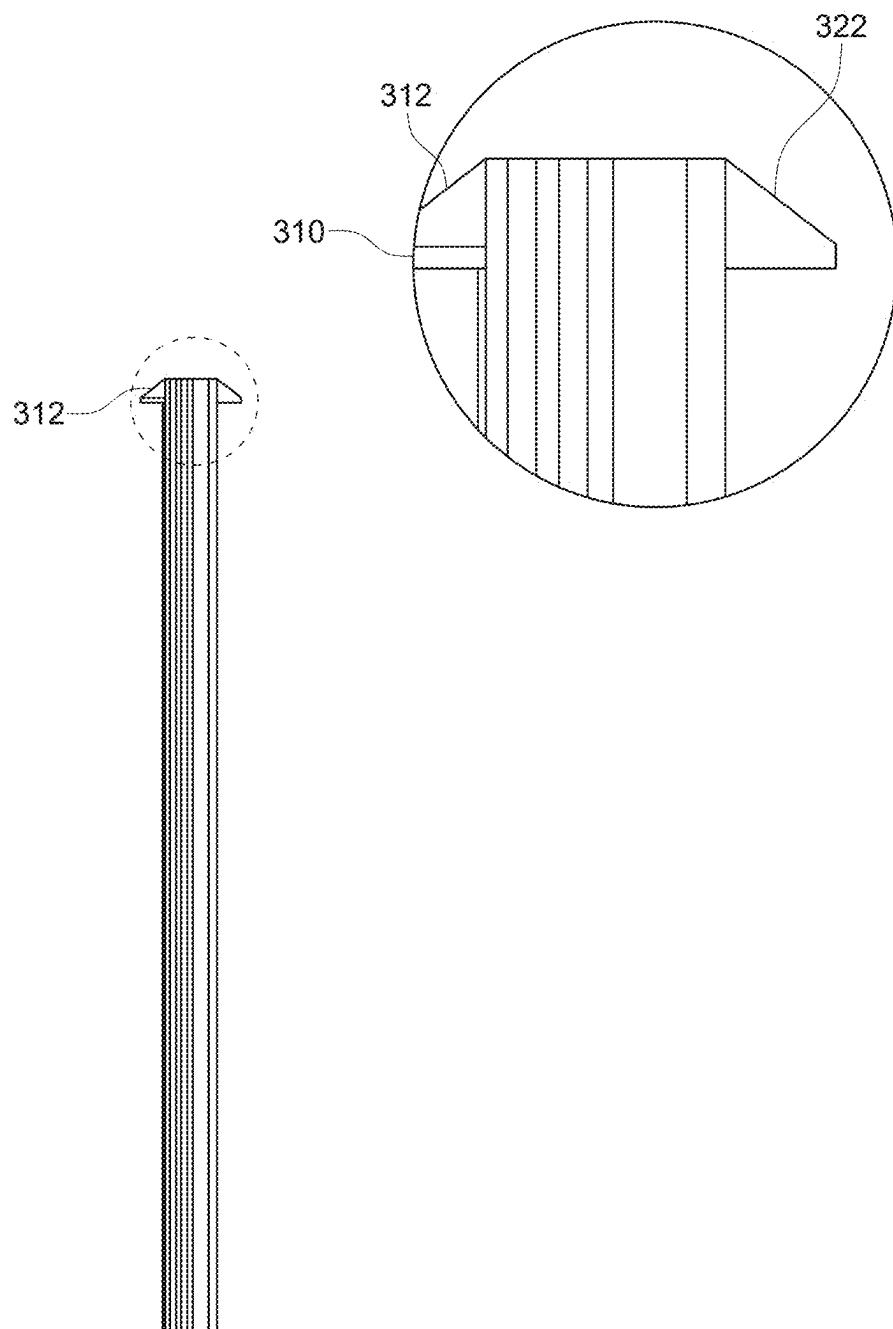


FIG. 11

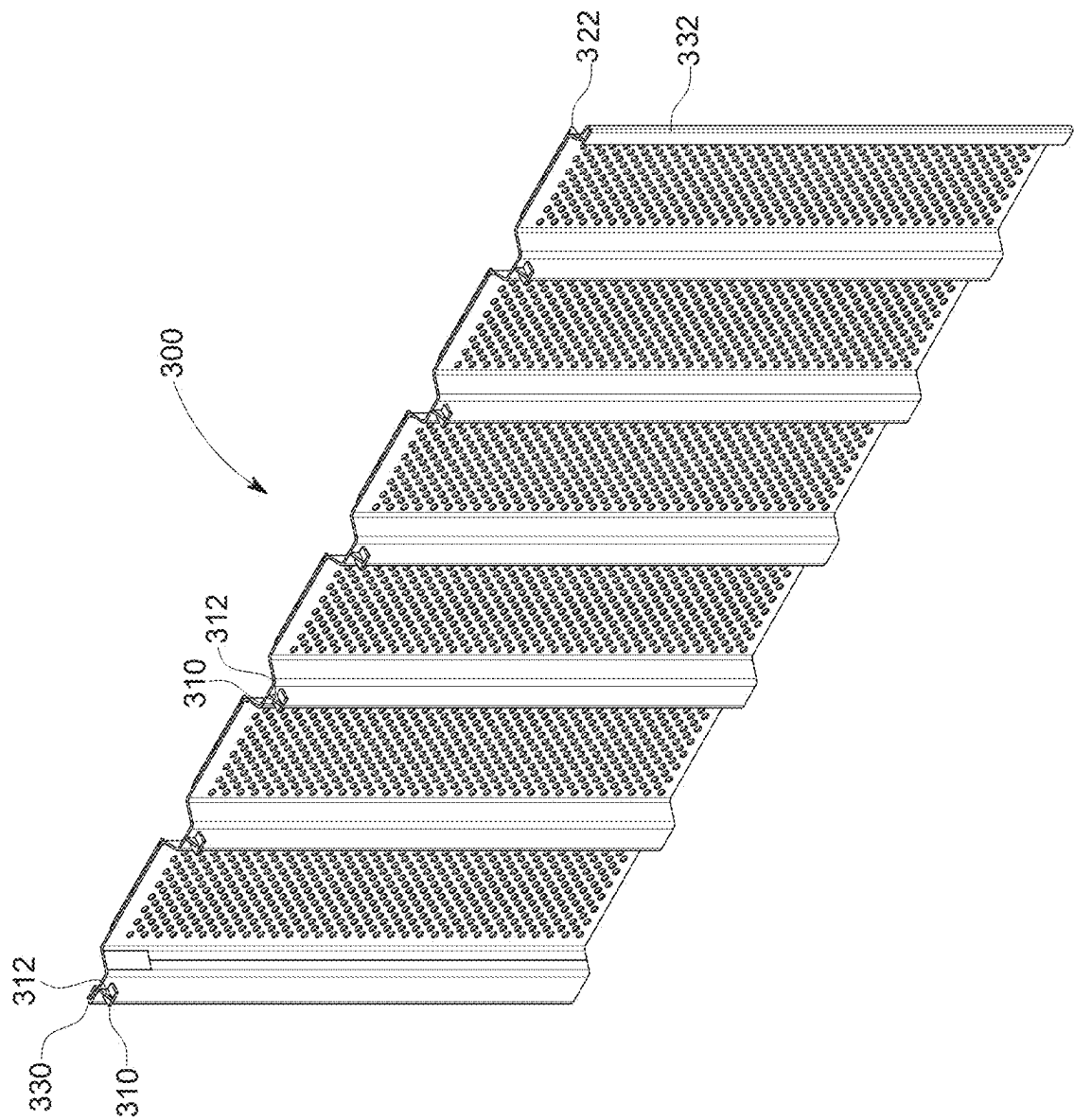


FIG. 12

LOCKING SOFFIT

BACKGROUND OF THE INVENTION

[0001] A soffit provides attic ventilation for homes and other structures, as well as a clean and neat surface beneath roof eaves and edges. The soffit covers the underside of a roof where it extends beyond the walls of a building and, therefore, the soffit covers a portion of an opening to an attic of the building. As heat builds up inside the attic, cooler air enters the attic through the soffit. Various types of vinyl or aluminum soffit panels and installation fixtures are regularly used, especially for wood-framed homes and structures.

[0002] The current art for installing a soffit utilizes channels such as J-channels, F-channels, and L-channels to secure the soffit panel between the fascia and the wall of the home or building. As the soffit panels are typically made from various types of vinyl or aluminum, the soffit panels are somewhat flexible and resilient. This flexibility and resiliency are useful in installing the soffit panels between channels in the fascia and wall as the soffit panel is often flexed to insert into the channels. Furthermore, the soffit panels are often formed or cut to a width that is slightly less than the distance between the inner edges of the channels to allow for uneven walls, uneven fascia, or other tolerances.

[0003] The current art works well for installation and during most situations, but when there is an extreme pressure difference between air in the attic and external air, there is significant air pressure and, therefore, air flow either into or out of the attic, going through the soffit. This significant air flow causes the soffit panels to flex and, if the soffit panels flex enough, the soffit panels disengage from one or both of the channels and the soffit panels usually fall to the ground. In such cases, if the soffit panel is not damaged and has not blown away, the soffit panel can be reinserted between the channels, but often this requires an installer and a ladder at considerable costs. Such loss of soffit panels happens often in severe weather such as hurricanes, straight-line winds, tornadoes, etc.

[0004] What is needed is a system that will keep the soffit panel in place during most air flow conditions.

SUMMARY OF THE INVENTION

[0005] In one embodiment, a system for securing the end of a soffit panel to a structure is disclosed. The system includes a channel for affixing to the structure (e.g., with adhesives, screws, nails). The channel has a locking mechanism. An end of the soffit panel has a barb/hook mechanism such that after inserting the barb/hook mechanism into the locking mechanism, the barb/hook mechanism and the locking mechanism hold that end of the soffit panel within the channel. The barb/hook mechanism is anticipated to be inwardly facing, or a pointed barb.

[0006] In another embodiment, a system for securing an end of a soffit panel to a structure is disclosed. The system includes a channel (e.g., an F-channel, U-channel, wall channel, track, wall track) that is affixed to the structure. The channel includes a mechanism for locking an end of the soffit panel that has an upper hook and a lower hook such that after inserting the end of the soffit panel into the mechanism for locking, the upper hook and the lower hook locks the end of the soffit panel within the channel.

[0007] In another embodiment, a soffit panel is disclosed. The soffit panel has alternating, offset upper surfaces and

lower surfaces. The soffit panel has a first lengthwise end for affixing to a fascia of a roof system and a second lengthwise end for insertion into a channel (e.g., an F-channel, U-channel, wall channel, track, wall track), the second lengthwise end has an upper edge that includes upper hooks for locking into the channel after the second lengthwise edge is inserted into the F-channel.

[0008] In another embodiment, a soffit system is disclosed. The soffit system includes a channel (e.g., an F-channel, U-channel, wall channel, track, wall track) for affixing to a structure. The channel has a base, a first leg, and a second leg. A first end of the first leg connects to the base at approximately 90 degrees and a first end of the second leg connects to the base at approximately 90 degrees. A second distal end of the first leg has a first catch and a second distal end of the second leg has a second catch. The first catch is on a side of the first leg that is towards the second leg and the second catch is on a side of the second leg that is towards the first leg. The soffit system includes a soffit panel that has a first lengthwise side and a second lengthwise side. The first lengthwise side has an upper hook and a lower hook for inserting the first lengthwise side into the channel between the first catch and the second catch, thereby locking the upper hook by the first catch and the lower hook by the second catch.

BRIEF DESCRIPTION OF DRAWINGS

[0009] The invention can be best understood by those having ordinary skill in the art by reference to the following detailed description when considered in conjunction with the accompanying drawings in which:

[0010] FIG. 1 illustrates a schematic view of a soffit installed to a building as per the prior art.

[0011] FIG. 2 illustrates a schematic view of a soffit installed to a building as per the prior art showing the effects of high-pressure gradients or wind.

[0012] FIGS. 3 and 3A illustrate schematic views of a first mode of soffit installed to a building as per the present invention.

[0013] FIGS. 4 and 4A illustrate schematic views of a second mode of soffit installed to a building as per the present invention.

[0014] FIGS. 5 and 5A illustrate schematic views of a third mode of soffit installed to a building as per the present invention.

[0015] FIG. 6 illustrates a detailed perspective view of the soffit channel shown in FIG. 5.

[0016] FIG. 7 illustrates a schematic view of a fourth mode of the soffit installed to a building as per the present invention.

[0017] FIG. 8 illustrates a detailed perspective view of the soffit channel shown in FIG. 7 before insertion of the soffit.

[0018] FIG. 8A illustrates a detailed perspective view of the soffit channel shown in FIG. 7 after insertion of the soffit.

[0019] FIG. 9 illustrates a side plan view of the fourth mode of soffit installed to a building as per the present invention.

[0020] FIG. 10 illustrates a top plan view of the fourth mode of soffit as per the present invention.

[0021] FIG. 11 illustrates an edge plan view of the fourth mode of soffit as per the present invention.

[0022] FIG. 12 illustrates a top perspective view of the fourth mode of soffit as per the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0023] Reference will now be made in detail to the presently preferred embodiments of the invention, examples of which are illustrated in the accompanying drawings. Throughout the following detailed description, the same reference numerals refer to the same elements in all figures.

[0024] Throughout this description, certain portions of the building such as the drip-edge, fascia, and wall construction are omitted for brevity and clarity reasons.

[0025] Referring to FIG. 1, a schematic view of a prior soffit panel **100** installed to a building as per the prior art is shown. In the prior art, each prior soffit panel **100** is installed at one end into a channel **102** and affixed to a board **112** of the fascia at the other end, for example, using a nail **105**. The channel **102** is affixed to the wall **116**, typically using fasteners such as screws or nails **105**. Although shown as F-channels in the drawings, there are many forms of channels in the prior art such as J-channels, F-channels, and L-channels, U-channels, wall cannels, tracks, wall tracks.

[0026] In FIG. 1, a portion of a roof **114** is also shown. Note that the prior soffit panel **100** is typically made of aluminum or a plastic such as vinyl and, being made of such, is stiff, yet bendable and resilient enough to bend for insertion into the channel **102**. Also note that the prior soffit panels **100** are typically cut shorter than the distance between the wall **116** and the board **112** of the fascia to account for uneven walls **116** and boards **112**.

[0027] In some embodiments, holes (not shown for brevity and clarity reasons) are formed along one edge of the prior soffit panels **100** of the prior art (not shown for brevity and clarity reasons) and the nails **105** are hammered through the holes to fasten the prior soffit panel **100** to the board **112** of the fascia. As the prior soffit panel **100** is typically made of sheet aluminum or vinyl, often with the day-to-day flexing caused by normal air flow, the prior soffit panel **100** weakens at these nail interfaces.

[0028] Referring to FIG. 2, a schematic view of a prior soffit installed to a building as per the prior art showing the effects of high-pressure gradients or wind is shown. In this view, the same prior soffit panel **100** was installed at one end into the channel **102** and affixed to the board **112** of the fascia at the other end using, for example, a nail **105**; the channel **102** affixed to the building. A pressure different between the attic and the ambient air has caused a strong air flow into the attic from the ambient air causing the prior soffit panel **100** to flex or bend so much that the end of the prior soffit panel **100** has left the channel **102** and, in this example, the prior soffit panel **100** will move into the attic space and, once the strong air flow abates, the prior soffit panel **100** will likely fall to the ground or blow away if it is windy as the prior soffit panel **100** is often very weak around where the nail **105** is placed.

[0029] Referring to FIGS. 3 and 3A, schematic views of a soffit panel **200** having a locking mechanism **201** are shown installed to a building as per the present invention. In some embodiments, the soffit panel **200** is installed between two locking channels **202** (as shown in FIG. 3) or in some embodiments the soffit panel **200** is installed between one locking channel **202** and affixed to the board **112** of the fascia at an opposite end using, for example, a nail **105** as shown in FIG. 3A. In the example of FIG. 5, one locking channel **202** is affixed to the board **112** of the fascia and the other locking channel **202** is affixed to the wall **116** of the building,

typically using fasteners such as screws or nails. Each locking channel **202** has an inwardly aimed barb **206** (e.g., an inwardly aimed barb **206** that is formed or attached at an end of the locking channel **202** and opens toward the surface to which the locking channel **202** is affixed (e.g., board **112** or wall **116**). The soffit panel **200** has a barb **204** at one or both ends. The barb **204** is formed or affixed to an end area of the soffit panel **200** and opening towards the opposing end of the soffit panel **200**. As the soffit panel **200** and/or the locking channel **202** are made from metal (e.g., aluminum, galvanized steel, galvalume) or plastic (e.g., vinyl), the inwardly aimed barb **206** and/or the barb **204** on the soffit panel **200** are resilient. Therefore, as the end of the soffit panel **200** is inserted into the locking channel **202**, one or both of the inwardly aimed barb **206** and/or the barb **204** on the soffit panel **200** bend, allowing for insertion. Once the barb **204** on the soffit panel **200** passes the inwardly aimed barb **206**, one or both of the inwardly aimed barb **206** and/or the barb **204** on the soffit panel **200** return to substantially their original shape as shown in FIG. 3. When a strong air flow occurs (as in FIG. 2), the inwardly aimed barb **206** catches on to the barb **204** on the soffit panel **200**, resisting release of the soffit panel **200** from the locking channel **202**. Note that in embodiments in which a single locking channel **202** is utilized (e.g., a nail **105** is used at one end of the soffit panel **200** as in FIG. 3A), it is anticipated that during strong air flows, the end of the soffit panel **200** that is held by the nail **105** may come free from the nail **105**, but the opposite end of the soffit panel **200** will remain in the locking channel **202** and not blow away or fall and possibly hurt somebody. Note that the loss of a prior soffit panel **100** during a storm allows water entry into the attic of the building.

[0030] For completeness, a portion of a roof **114** is also shown. Note that the soffit panel **200** is typically made of metal (e.g., aluminum, galvanized steel, galvalume) or plastic (e.g., vinyl) and, being made of such, is stiff, yet bendable and resilient enough to bend for insertion between the locking channels **202** or between a locking channel **202** and a channel **102** of the prior art. Also note that the soffit panels **200** are typically cut shorter than the distance between the wall **116** and the board **112** to account for uneven walls **116** and board **112**.

[0031] Referring to FIGS. 4 and 4A, schematic views of a soffit installed to a building as per the present invention using a locking mechanism **201A** are shown. In one embodiment, as shown in FIG. 4, the soffit panel **200** is installed between two locking channels **202A** while in some embodiments as shown in FIG. 4A, the soffit panel **200** is held at one end by a single locking channel **202A** and the other end is affixed to the board **112** (e.g., by a nail **105**). In the embodiment of FIG. 5, one locking channel **202A** is affixed to the board **112** of the fascia and the other locking channel **202A** is affixed to the wall **116** of the building, typically using fasteners such as screws or nails. Each locking channel **202A** has an inwardly aiming step **206A** (e.g., an inwardly aiming step **206A** that is formed or attached at an end of the locking channel **202A** and increases in width toward the surface (e.g., board **112** or wall **116**) to which the locking channel **202A** is affixed. The soffit panel **200** has a barb **204** at one or both ends, formed or affixed to an end area of the soffit panel **200** and opening towards the opposing end of the soffit panel **200**. As the soffit panel **200** and/or the locking channel **202A** are made using metal (e.g., aluminum, galvanized steel, galvalume) or plastic (e.g., vinyl), the

inwardly aiming step **206A** and/or the barb **204** on the soffit panel **200** are resilient. Therefore, as the end of the soffit panel **200** is inserted into the locking channel **202A**, one or both of the inwardly aiming step **206A** and/or the barb **204** on the soffit panel **200** will bend or deform, allowing for insertion. Once the barb **204** on the soffit panel **200** passes the inwardly aiming step **206A**, one or both of the inwardly aiming step **206A** and/or the barb **204** on the soffit panel **200** return to substantially their original shape as shown in FIG. 4. When a strong air flow occurs (as in FIG. 2), the barb **204** catches on to the inwardly aiming step **206A**, resisting release of the soffit panel **200** from the locking channel **202A**. Note that in embodiments in which a single locking channel **202A** is utilized (e.g., a nail **105** is used to secure one end of the soffit panel **200** to the board **112** of the fascia), it is anticipated that during strong air flows, the end of the soffit panel **200** that is nailed to the board **112** might fail, but the soffit panel **200** will remain in the locking channel **202A** and not blow away or fall and possibly hurt somebody.

[0032] For completeness, a portion of a roof **114** is also shown. Note that the soffit panel **200** is typically made of metal (e.g., aluminum, galvanized steel, galvalume) or plastic (e.g., vinyl), and, being made of such, is stiff, yet bendable and resilient enough to bend for insertion between the locking channels **202A** or between a locking channel **202A** and a channel **102** of the prior art. Also note that the soffit panels **200** are typically cut shorter than the distance between the wall **116** and the board **112** to account for uneven walls **116** and board **112**.

[0033] Referring to FIGS. 5 and 5A, schematic views of a soffit panel **200B** having a locking mechanism **201B** is shown installed to a building as per the present invention. In some embodiments, as shown in FIG. 5, the soffit panel **200B** is installed between two locking channels **202B** while in some embodiments, the soffit panel **200B** is installed using a locking channel **202B** at one end (e.g., at the wall **116** of the building) and affixed to the board **112** of the fascia at the other end (e.g., using nails **105**). In the embodiment of FIG. 5, one locking channel **202B** is affixed to the board **112** and the other locking channel **202B** is affixed to the wall **116** of the building, typically using fasteners such as screws or nails. Each locking channel **202B** has notched orifice **206B** (e.g. formed or affixed to an inside surface of the locking channel **202B** with a notch **207B**). The soffit panel **200B** has a pointed barb **208B** at one or both ends, formed or affixed to an end area of the soffit panel **200B** and pointing outwardly from the soffit panel **200**. The pointed barb **208B** has a catch edge **204B** that is formed to catch on the notch **207B** after insertion of the pointed barb **208B** into the locking channel **202B**.

[0034] As the soffit panel **200B** and/or the locking channel **202B** are made using metal (e.g., aluminum, galvanized steel, galvalume) or plastic (e.g., vinyl), the pointed barb **208B** and/or the notched orifice **206B** are resilient. Therefore, as the pointed barb **208B** is inserted into the locking channel **202B**, an angled edge **209B** (see FIG. 6) of the pointed barb **208B** on the soffit panel **200** causes the notched orifice **206B** to bend or deform, allowing for insertion of the pointed barb **208B**. Once the pointed barb **208B** on the soffit panel **200** passes the notched orifice **206B**, the notched orifice **206B** returns to substantially its original shape as shown in FIG. 5. When a strong air flow occurs (as in FIG. 2), the pointed barb **208B** catches on to the notch **207B**, resisting release of the soffit panel **200B** from the locking

channel **202B**. Note that in embodiments in which a single locking channel **202B** is utilized (e.g., as in FIG. 5A, one end of the soffit panel is affixed to the board **112** using a nail **105**), it is anticipated that during strong air flows, the end of the soffit panel that is affixed to the board **112** (e.g., using a nail **105**), the connection may fail, but the soffit panel **200B** will remain in the locking channel **202B** and not blow away or fall and possibly hurt somebody.

[0035] For completeness, a portion of a roof **114** is also shown. Note that the soffit panel **200B** is typically made of metal (e.g., aluminum, galvanized steel, galvalume) or plastic (e.g., vinyl), and, being made of such, is stiff, yet bendable and resilient enough to bend for insertion between the locking channels **202B** or between a locking channel **202B** and a channel **102** of the prior art. Also note that the soffit panels **200B** are typically cut shorter than the distance between the wall **116** of the building and the inside surface of the board **112** to account for uneven walls **116** of the building and board **112**.

[0036] Referring to FIG. 6, a detailed perspective view of one embodiment of the locking soffit channel **202B** shown in FIG. 5. In some embodiments, the locking soffit channel **202B** is fabricated as a single, elongated section, for example, extruded metal (e.g., aluminum, galvanized steel, galvalume) or plastic (e.g., vinyl).

[0037] Although the soffit panels **200/200B** are shown with barbs **204** or pointed barbs **208B** at both ends of the soffit panel **200/200B**, it is equally anticipated that one end of the soffit panels **200/200B** have no locking mechanism for interfacing with a channel **102** or other mounting mechanism such as using a nail **105**. Further, it is fully anticipated that any combination of locking mechanisms be provided on each soffit panel **200/200B**.

[0038] Referring to FIG. 7, a schematic view of a fourth mode of soffit installed onto a building as per the present invention is shown. In the embodiment shown, the soffit panel **300** is installed into a locking F-channel **400** and affixed to the board **112** of the fascia at the other end of the soffit panel **300** (e.g., using fasteners such as nails **105**). Note that although an F-channel **400** is shown, any form/shape of channel (e.g., an F-channel, U-channel, wall channel, track, wall track) is fully anticipated.

[0039] The locking F-channel **400** is affixed to the wall **116** of the building, typically using fasteners such as screws or nails **105**. Each locking F-channel **400** has a base **405** that has at least one orifice **406** (see FIGS. 8 and 8A) through which the fastener or nail **105** passes for holding the locking F-channel **400** to the wall **116**. The locking F-channel **400** has two legs **402** that extend perpendicularly (90 degrees) or approximately perpendicularly (approximately 90 degrees) at a first end from the base **405** to form an “F” shape. At the other end of each leg **402** is a catch **404** that captures the hooks **312/322** of the soffit panel **300** (shown captured in FIG. 7). Once the hooks **312/322** of the soffit panel **300** enter the locking F-Channel **400**, it is difficult for the soffit panel **300** to exit the locking F-channel **400**, thereby resisting from being dislodged during strong winds.

[0040] As the soffit panel **300** and/or the locking F-channel **400** are made using metal (e.g., aluminum, galvanized steel, galvalume) or plastic (e.g., vinyl, polyvinyl chloride). The legs **402** are resilient to allow the legs **402** and catches **404** to spread apart during insertion of the soffit panel **300**, then closing back to lock the hooks **312/322** of the soffit panel **300** behind the catches **404**. Therefore, as the hooks

312/322 are inserted into the locking F-channel 400, the barbs 312/322 on the soffit panel 300 causes the legs 402 and catches 404 to bend or deform, allowing for insertion of the hooks 312/322. Once the hooks 312/322 on the soffit panel 300 pass the catches 404, the legs 402 and catches 404 return to substantially their original shape as shown in FIG. 7. When a strong air flow occurs (as in FIG. 2), the hooks 312/322 cannot easily escape the catches 404, resisting release of the soffit panel 300 from the locking F-channel 400. Note that it is anticipated that during strong air flows, the end of the soffit panel that is affixed to the board 112 (e.g., using a nail 105) may fail and come loose, but the soffit panel 300 will likely remain in the locking F-channel 400 and not blow away or fall and possibly hurt somebody.

[0041] For completeness, a portion of a roof 114 is also shown. Note that the soffit panel 300 and the locking F-channel is typically made of metal (e.g., aluminum, galvanized steel, galvalume) or plastic (e.g., vinyl, polyvinyl chloride).

[0042] Referring to FIGS. 8 and 8A, detailed perspective views of the locking F-channel 400 of FIG. 7 are shown before insertion of the soffit panel 300 (FIG. 8) and after insertion of the soffit panel 300 (FIG. 8A). A side cut-away view of the locking F-channel 400 is shown. Typically, the locking F-channel 400 is affixed to the wall 116 before insertion of the soffit panel 300. In such, the locking F-channel 400 is held to the wall 116 by a fastener or nail 105 that passes through an orifice 406 in the base 405 of the locking F-channel 400 and into the wall 116, though any mounting scheme is anticipated including the use of an adhesive or double-sided tape.

[0043] As described, the locking F-channel 400 has two legs 402 that extend perpendicularly or approximately perpendicularly at a first end from the base 405 to form an “F” shape. At the other end of each leg 402 is a catch 404 that captures the barbs 312/322 of the soffit panel 300 (shown captured in FIGS. 7 and 8A). Once the hooks 312/322 of the soffit panel 300 enter the locking F-channel 400, it is difficult for the soffit panel 300 to exit the locking F-channel 400, thereby resisting from being dislodged during strong winds.

[0044] As the soffit panel 300 and/or the locking F-channel 400 are made using metal (e.g., aluminum, galvanized steel, galvalume) or plastic (e.g., vinyl, polyvinyl chloride). The legs 402 are resilient to allow the legs 402 to spread apart during insertion of the soffit panel 300, then closing back to lock the hooks 312/322 of the soffit panel 300 behind the catches 404. Therefore, as the hooks 312/322 are inserted into the locking F-channel 400, the hooks 312/322 on the soffit panel 300 causes the legs 402 and catches 404 to bend or deform, allowing for insertion of the hooks 312/322. Once the hooks 312/322 on the soffit panel 300 pass the catches 404, the legs 402 and catches 404 return to substantially their original shape as shown in FIGS. 7 and 8A. When a strong air flow occurs (as in FIG. 2), the hooks 312/322 cannot easily escape the catches 404, resisting release of the soffit panel 300 from the locking F-channel 400.

[0045] Referring to FIGS. 9, 10, 11, and 12 views of the fourth mode of soffit panel 300 as per the present invention are shown. In FIG. 9, the front of the upper hook 310 is shown with upper angled steps 312. Note that upper direction is generally the direction of the upper surface 316 of the soffit panel 300 when installed horizontally between a building and a fascia board, though any configuration of

installation is anticipated. Although three upper angled steps 312 are shown for each upper hook 310, any number of upper angled steps 312 are anticipated, including one upper angled step 312 that is partially or the full length of the upper hook 310. The upper angled steps 312 engage with one of the catches 404 of the locking F-channel 400 to displace that leg 402 and catch 404 of the locking F-channel 400 as the soffit panel 300 is inserted into the locking F-channel 400. After the upper angled steps 312 pass into the locking F-channel 400 (beyond the catch 404), the leg 402/catch 404 returns to approximately that leg 402/catch 404 original shape, locking the upper hook 310 against the catch 404.

[0046] In FIG. 9, the front of the lower hook 320 is shown with lower angled steps 322. Although one lower angled step 322 is shown for each lower hook 320, any number of lower angled steps 322 are anticipated, including one lower angled step 322 that is partially or the full length of the lower hook 320. The lower angled step 322 engages with the other leg 402 and catch 404 of the locking F-channel 400 to displace that leg 402 and catch 404 of the locking F-channel 400 as the soffit panel 300 is inserted into the locking F-channel 400. After the lower angled steps 322 pass into the locking F-channel 400 (beyond the catch 404), the leg 402 and catch 404 return to approximately that leg 402 and catch 404 original shape, locking the lower hook 320 against the catch 404.

[0047] In some embodiments, for rigidity, the soffit panel 300 has an upper surface 316 that is offset from a lower surface 326. In some embodiments, one or both of the upper surface 316 and lower surface 326 includes a plurality of orifices while in some embodiments, the upper surface 316 and the lower surface 326 are solid without orifices.

[0048] At each end of the soffit panel 300 is a set of hooks 330/332. During installation, the left hook 330 of the hooks 330/332 of a first soffit panel 300 is interfaced to a right hook 332 of the hooks 330/332 of an adjacent soffit panel 300 to hold the first soffit panel 300 to the adjacent soffit panel, etc.

[0049] In FIG. 10, the surface of the upper surface 316 and lower surface 326 is shown. Note that the lower surface 326 is shown having air vent holes, though in some embodiments, the soffit panel is without air vent holes or having air vent holes in the upper surface 316.

[0050] In FIG. 11, an edge view is shown. Note that the upper hook 310 is shown with upper angled steps 312 as the upper angled step 312 is set back from the edge of the upper hook 310 as shown in FIG. 9, yet the lower hook 320 appears solid as the lower angled step 322 is shown at the end of the lower hook 320.

[0051] A perspective view of the soffit panel 300 is shown in FIG. 12. The upper hooks 310 and upper angled steps 312 are shown and only part of the lower angled steps 322 are visible in this view.

[0052] Equivalent elements can be substituted for the ones set forth above such that they perform in substantially the same manner in substantially the same way for achieving substantially the same result.

[0053] It is believed that the system and method as described and many of its attendant advantages will be understood by the foregoing description. It is also believed that it will be apparent that various changes may be made in the form, construction, and arrangement of the components thereof without departing from the scope and spirit of the invention or without sacrificing all of its material advantages. The form herein before described being merely exem-

plary and explanatory embodiment thereof. It is the intention of the following claims to encompass and include such changes.

1. A system for securing an end of a soffit panel to a structure, the system comprising:

a channel that is affixed to the structure, the channel having means for locking; and

the end of the soffit panel having an upper hook and a lower hook such that after inserting the end of the soffit panel into the means for locking, the upper hook and the lower hook locks the end of the soffit panel within the channel.

2. The system for securing the end of the soffit panel to the structure of claim 1, whereas the channel comprises a base, the base has an orifice for accepting a fastener when the channel is affixed to a building and the means for locking comprises a first leg comprising a first catch and a second leg comprising a second catch.

3. The system for securing the end of the soffit panel to the structure of claim 2, whereas the first leg extends from the base at an approximately 90 degree angle and the second leg extends from the base at the approximately 90 degree angle.

4. The system for securing the end of the soffit panel to the structure of claim 2, whereas the first catch extends from the first leg at an approximately 90 degree angle and the second catch extends from the second leg at the approximately 90 degree angle, and the first catch is positioned towards the second catch and spaced apart from the second catch.

5. The system for securing the end of the soffit panel to the structure of claim 2, whereas the upper hook comprises upper angled step for displacing the first catch and first leg.

6. The system for securing the end of the soffit panel to the structure of claim 2, whereas the lower hook comprises lower angled step for displacing the second catch of the second leg.

7. The system for securing the end of the soffit panel to the structure of claim 1, wherein the soffit panel and/or the channel comprise aluminum.

8. The system for securing the end of the soffit panel to the structure of claim 1, wherein the soffit panel and/or the channel comprise plastic.

9. The system for securing the end of the soffit panel to the structure of claim 8, wherein the plastic is vinyl.

10. A soffit panel, the soffit panel comprising:

the soffit panel has alternating, offset upper surfaces and lower surfaces; and

the soffit panel has a first lengthwise end for affixing to a fascia of a roof system and a second lengthwise end for insertion into an channel, the second lengthwise end has an upper edge that includes upper hooks.

11. The soffit panel of claim 10, whereas a lower edge of the second lengthwise end includes lower hooks.

12. The soffit panel of claim 10, whereas a first non-lengthwise end has a left hook and a distal second non-lengthwise end has a right hook for connecting adjacent soffit panels to each other.

13. The soffit panel of claim 10, wherein the upper surfaces have orifices.

14. A soffit system comprising:

a channel for affixing to a structure, the channel having a base, a first leg, and a second leg, a first end of the first leg connects to the base at approximately 90 degrees and a first end of the second leg connects to the base at the approximately 90 degrees, a second distal end of the first leg has a first catch and the second distal end of the second leg has a second catch, the first catch is on a side of the first leg that is towards the second leg and the second catch is on a side of the second leg that is towards the first leg; and

a soffit panel having a first lengthwise side and a second lengthwise side, the first lengthwise side having an upper hook and a lower hook for inserting the first lengthwise side into the channel between the first catch and the second catch for locking the upper hook by the first catch and the lower hook by the second catch.

15. The soffit system of claim 14, whereas the first catch meets the first leg at the approximately 90 degrees and the second catch meets the second leg at the approximately 90 degrees.

16. The soffit system of claim 14, whereas the upper hook comprises upper angled step for displacing the first catch and the first leg.

17. The soffit system of claim 14, whereas the lower hook comprises lower angled step for displacing the second catch and the second leg.

18. The soffit system of claim 14, wherein the base of the channel includes an orifice for accepting a fastener for affixing the channel to a structure.

19. The soffit system of claim 14, wherein the soffit panel and/or the channel comprise aluminum.

20. The soffit system of claim 14, wherein the soffit panel and/or the channel comprise plastic.

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